

ECORYX SYSTEM ARCHITECTURE & API MANUAL

AI-Powered Geospatial Environmental Clearance Platform // Technical Guide

Official Palette: Warm Off-White, Charcoal, Muted Forest Green, Amber (SIH 2026)

1. End-to-End System Pipeline Architecture

[Step 1: User Input Layer]

Latitude, Longitude, Project Footprint (Ha), Search Radius (km), Industry Category

[Step 2: FastAPI Data Aggregation Engine]

High-Throughput Asynchronous Orchestrator (<0.38s parallel multi-source ingestion)

Concurrent Ingestion Services (Step 3):

1. Weather & AQI APIs

PM2.5, PM10, AQI Index, Temperature, Wind direction & speed

2. Overpass (OSM)

River drainage basins, water bodies, wetlands, reserve forests

3. PostGIS DB

R-Tree spatial index, Wildlife Sanctuaries, IUCN Red List status

4. Demographics

Buffer population density, nearest village/habitation clusters

[Step 4: Compiled Site Environmental JSON]

Consolidated standardized GeoJSON payload ready for statutory cross-referencing

[Step 5: Gemini API + Regulatory Rulebook]

2,400+ EIA Gazette Laws checked, Sensitivity Score (0-100), Environmental Management Plan (EMP)

[Step 6: PDF Report Engine (ReportLab)]

Generates finalized, certified, audit-ready EIA Clearance PDF with digital verification seal

2. API & Data Services Specifications

FastAPI Data Aggregation Core

Endpoint/Tech: POST /api/v1/clearance/audit

Receives user coordinates, dispatches asynchronous asyncio coroutines to query PostGIS, Overpass, Weather, and Demographics in < 0.38s, and produces consolidated JSON.

Code Location: src/components/ConsoleView/ConsoleSection.jsx & src/components/ArchitectureSection.jsx

Weather & AQI Atmospheric Monitor

Endpoint/Tech: GET https://air-quality-api.open-meteo.com/v1/air-quality

Retrieves PM2.5, PM10, AQI Index, Ambient Temperature (°C), and Wind Direction for air pollution dispersion radius modeling.

Code Location: src/constants/presets.js (weather_aqi schema) & src/components/ConsoleView/ScorecardTab.jsx

Overpass API (OpenStreetMap)

Endpoint/Tech: POST https://overpass-api.de/api/interpreter

Performs spatial bounding radius queries to detect freshwater streams, river basins, canals, wetlands, and reserve forest boundaries.

Code Location: src/components/ConsoleView/SpatialTab.jsx (Hydro & Landuse Layers)

PostGIS Spatial Engine

Endpoint/Tech: PostgreSQL 16 + PostGIS 3.4 (Internal Database)

Executes ST_DWithin & ST_Intersects using R-Tree spatial indexing across 106 National Parks, 573 Wildlife Sanctuaries, and IUCN Red List species habitats.

Code Location: src/constants/presets.js (postgis_db schema) & src/components/ConsoleView/ScorecardTab.jsx

Demographics Engine

Endpoint/Tech: Local Spatial Census Boundary Service

Calculates the resident human population residing within the project radial buffer and measures distance to nearest village/habitation.

Code Location: src/components/ConsoleView/ScorecardTab.jsx & src/constants/presets.js

Gemini API + Regulatory Rulebook

Endpoint/Tech: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-1.5-pro:generateContent

Cross-checks 2,400+ digitized statutory EIA laws (WPA 1972, FCA 1980, Water Act 1974, Air Act 1981, CRZ 2019), computes 0-100 Sensitivity Score, and drafts EMP.

Code Location: src/components/ConsoleView/ScorecardTab.jsx & src/components/ConsoleView/EmpTab.jsx

PDF Report Engine (ReportLab)

Endpoint/Tech: Client-side jsPDF & Backend Python ReportLab

Compiles certified, audit-ready EIA Screening PDF reports with radar charts, spatial tables, and official mitigation action plans.

Code Location: src/utlis/exportPDF.js & src/components/ConsoleView/ConsoleSection.jsx

3. Official Color Palette & Artwork Themes

Exact tokens configured across `src/index.css` and `src/context/ThemeContext.jsx`:

- 

Background (Warm Off-White) — #f3f1eb
Main page background canvas and backdrop tint
- 

Surface (Warm Light Card) — #fbfaf6
Card containers, UI panels, elevated white widgets
- 

Dark Surface (Charcoal) — #222a25
Dark hero console bar, raw code preview, high-contrast dark buttons
- 

Primary (Muted Forest Green) — #315c48
Primary CTA buttons, brand badges, active GIS layers, headlines
- 

Primary Dark — #244737
Primary button hover states, deep contrast typography
- 

Amber (Charging States & Caution) — #b77927
Active charging states, solar energy, caution notifications
- 

Red (Alerts) — #a54d42
Protected area violations, air quality warnings, old manual process alerts
- 

Text (Charcoal Body) — #20231f
High-contrast readable headings and body copy
- 

Muted Text — #73766f
Subtitles, secondary labels, coordinate metadata
- 

Borders (Warm Subtle) — #d8d4ca
Card borders, table dividers, input bounding lines

4. Codebase File Structure & Navigation Map

src/App.jsx Main page layout orchestrator, state manager for coordinates, presets & audits
src/main.jsx App entry point wrapping root with ThemeProvider
src/index.css Global CSS variables (warm off-white, forest green, charcoal), rotary switch styles
src/context/ThemeContext.jsx Dynamic color theme engine with warm forest palette & localStorage persistence
src/components/ThemeSelector.jsx Color sample dropdown switcher with live preview in navbar
src/constants/presets.js Scenario presets with 4-stream architecture data (Aravalli, Ghats, Solar, etc.)
src/components/Header.jsx Top navigation bar with embedded ThemeSelector dropdown
src/components/Hero.jsx Hero section with TopographyRadiusMap, GIS stats, and meadow background
src/components/TopographyRadiusMap.jsx Interactive Leaflet GIS radius scanner with elevation contours & conflict pins
src/components/RotarySwitch.jsx Skeuomorphic 2-state rotary dial switching Before vs. After Ecoryx view
src/components/WorkbenchSection.jsx Main comparison container holding rotary switch & views
src/components/BeforeEcoryxView.jsx Breakdown of old manual paperwork delays, metric cards, & pain points
src/components/WorkflowSection.jsx 4-Step automated clearance workflow + Blume-inspired organic cloud quote card
src/components/SectorsSection.jsx Bento grid showing Solar, Transport, Hydrology & Statutory Legal intelligence
src/components/ArchitectureSection.jsx Live interactive architecture flowchart matching system pipeline diagram
src/components/CalculatorSection.jsx Interactive ROI calculator calculating review months & crore rupees preserved
src/components/Footer.jsx Footer with system status indicator & Smart India Hackathon 2026 credits
src/components/ConsoleView/ConsoleSection.jsx Clearance workspace with preset selector, tab navigation, & PDF export
src/components/ConsoleView/AuditForm.jsx Form inputs for Latitude, Longitude, Project Area (Ha), & Industry Category
src/components/ConsoleView/ScorecardTab.jsx 4 Ingestion Stream cards (PostGIS, OSM, AQL, Demographics) + Gemini verdict
src/components/ConsoleView/SpatialTab.jsx PostGIS & Overpass OSM spatial layers table with statutory rules & mitigation

src/components/ConsoleView/EmpTab.jsx

Official Action Plan: Clean Air filters, Zero Liquid Discharge & tree planting

src/components/ConsoleView/JsonTab.jsx

Consolidated site environmental JSON schema viewer with 1-click copy

src/components/ConsoleView/RadarChart.jsx

Chart.js 5-dimension vulnerability radar visualization

src/utls/exportPDF.js

Generates client-side certified EIA audit PDF deliverable

5. Stored Reference Images Directory

All architecture diagrams, color palettes, and visual design assets are stored in docs/reference_images/:

[1] docs/reference_images/01_architecture_diagram.png

Official Architecture Flowchart (88.5 KB)

Pipeline: User Input -> FastAPI -> 4 Ingestion Services (Weather/AQI, Overpass OSM, PostGIS, Demographics) -> JSON -> Gemini API -> ReportLab PDF

[2] docs/reference_images/02_blume_meadow_design_ref.jpg

Blume Meadow Design Benchmark (185.2 KB)

Inspiration visual showing lush rolling green hills, wild daisies, pure white cards, and organic cloud-shaped callout containers

[3] docs/reference_images/03_lush_meadow_hill_bg.jpg

Lush Meadow Hill Background Asset (1.08 MB)

High-definition rolling green grass hill background photograph with sunlit daisies used in the Hero section and background layers

[4] docs/reference_images/04_color_palette_ref_1.jpg

Artwork Reference 1 (Golden Rapeseed & Sky) (386.6 KB)

Cyclists riding through golden yellow rapeseed fields under azure blue sky (#D97706, #0284C7, #FAF7EE)

[5] docs/reference_images/05_color_palette_ref_2.jpg

Artwork Reference 2 (Canola Meadow & Red Accent) (183.5 KB)

Field of hope road through lush yellow blooms with red roof accent (#CA8A04, #65A30D, #DC2626)

[6] docs/reference_images/06_color_palette_ref_3.jpg

Artwork Reference 3 (Studio Ghibli Countryside) (209.1 KB)

Tabineko calendar scene with rice paddy greens, azure stream, and terracotta accents (#15803D, #0284C7, #FB923C)

[7] docs/reference_images/07_color_palette_ref_4.jpg

Artwork Reference 4 (Cool Summer Lagoon) (211.0 KB)

Emerald swimming cove with deep teal water and fresh lime foliage (#047857, #65A30D, #F5FAF7)

[8] docs/reference_images/08_color_palette_ref_5.jpg

Artwork Reference 5 (Minimalist Mustard & Olive) (106.6 KB)

Mustard field hillside with olive trees and earthy rustic textures (#B45309, #4D7C0F, #DC2626)